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SKILL 5

The queries demonstrate how SQL can retrieve, filter, join, group, sort, and analyze data using joins, operators, subqueries, and aggregate functions.

```
mysql> USE sec_13;
Database changed
mysql>
mysql> CREATE TABLE physician (
->     employeeid INT PRIMARY KEY,
->     name VARCHAR(100),
->     position VARCHAR(100),
->     ssn VARCHAR(20)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> INSERT INTO physician VALUES
-> (1, 'John Dorian', 'Staff Internist', '111111111'),
-> (2, 'Elliot Reid', 'Attending Physician', '222222222'),
-> (3, 'Christopher Turk', 'Surgical Attending Physician', '333333333'),
-> (4, 'Percival Cox', 'Senior Attending Physician', '444444444'),
-> (5, 'Bob Kelso', 'Head Chief of Medicine', '555555555'),
-> (6, 'Todd Quinlan', 'Surgical Attending Physician', '666666666'),
-> (7, 'John Wen', 'Surgical Attending Physician', '777777777'),
-> (8, 'Keith Dudemeister', 'MD Resident', '888888888'),
-> (9, 'Molly Clock', 'Attending Psychiatrist', '999999999');
Query OK, 9 rows affected (0.01 sec)
Records: 9  Duplicates: 0  Warnings: 0
```

```
mysql> CREATE TABLE department (  
->     departmentid INT PRIMARY KEY,  
->     name VARCHAR(100),  
->     head INT  
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> INSERT INTO department VALUES  
-> (1, 'General Medicine', 4),  
-> (2, 'Surgery', 7),  
-> (3, 'Psychiatry', 9);
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> show tables;
```

```
+-----+  
| Tables_in_sec_13 |  
+-----+  
| class             |  
| class_info        |  
| department         |  
| first_table       |  
| physician          |  
| second_table      |  
| student            |  
| student_marks1    |  
+-----+
```

8 rows in set (0.01 sec)

```
mysql> SELECT d.name AS Department,
->         p.name AS Head_Physician
-> FROM department d
-> JOIN physician p
-> ON d.head = p.employeeid;
```

Department	Head_Physician
General Medicine	Percival Cox
Surgery	John Wen
Psychiatry	Molly Clock

3 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM physician
-> WHERE position='Surgical Attending Physician';
```

employeeid	name	position	ssn
3	Christopher Turk	Surgical Attending Physician	333333333
6	Todd Quinlan	Surgical Attending Physician	666666666
7	John Wen	Surgical Attending Physician	777777777

3 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM physician
-> WHERE name LIKE 'John%';
```

employeeid	name	position	ssn
1	John Dorian	Staff Internist	111111111
7	John Wen	Surgical Attending Physician	777777777

2 rows in set (0.00 sec)

```
mysql> SELECT COUNT(*) AS Total_Physicians
-> FROM physician;
```

Total_Physicians
9

1 row in set (0.01 sec)

```
mysql> SELECT COUNT(DISTINCT position)
-> FROM physician;
```

COUNT(DISTINCT position)
7

1 row in set (0.00 sec)

```
mysql> SELECT position,
->      COUNT(*) AS Total_Employees
-> FROM physician
-> GROUP BY position;
```

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

7 rows in set (0.00 sec)

```
mysql> SELECT position,
->      COUNT(*) AS Total_Employees
-> FROM physician
-> GROUP BY position
-> HAVING COUNT(*) > 1;
```

position	Total_Employees
Surgical Attending Physician	3

1 row in set (0.00 sec)

```
mysql> SELECT *
-> FROM physician
-> ORDER BY name ASC;
```

employeeid	name	position	ssn
5	Bob Kelso	Head Chief of Medicine	555555555
3	Christopher Turk	Surgical Attending Physician	333333333
2	Elliot Reid	Attending Physician	222222222
1	John Dorian	Staff Internist	111111111
7	John Wen	Surgical Attending Physician	777777777
8	Keith Dudemeister	MD Resident	888888888
9	Molly Clock	Attending Psychiatrist	999999999
4	Percival Cox	Senior Attending Physician	444444444
6	Todd Quinlan	Surgical Attending Physician	666666666

9 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM physician
-> ORDER BY employeeid DESC;
```

employeeid	name	position	ssn
9	Molly Clock	Attending Psychiatrist	999999999
8	Keith Dudemeister	MD Resident	888888888
7	John Wen	Surgical Attending Physician	777777777
6	Todd Quinlan	Surgical Attending Physician	666666666
5	Bob Kelso	Head Chief of Medicine	555555555
4	Percival Cox	Senior Attending Physician	444444444
3	Christopher Turk	Surgical Attending Physician	333333333
2	Elliot Reid	Attending Physician	222222222
1	John Dorian	Staff Internist	111111111

9 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM physician
-> WHERE employeeid IN
-> (
-> SELECT head
-> FROM department
-> );
```

employeeid	name	position	ssn
4	Percival Cox	Senior Attending Physician	444444444
7	John Wen	Surgical Attending Physician	777777777
9	Molly Clock	Attending Psychiatrist	999999999

3 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM physician
-> WHERE employeeid NOT IN
-> (
-> SELECT head
-> FROM department
-> );
```

employeeid	name	position	ssn
1	John Dorian	Staff Internist	111111111
2	Elliot Reid	Attending Physician	222222222
3	Christopher Turk	Surgical Attending Physician	333333333
5	Bob Kelso	Head Chief of Medicine	555555555
6	Todd Quinlan	Surgical Attending Physician	666666666
8	Keith Dudemeister	MD Resident	888888888

6 rows in set (0.01 sec)

```
mysql> SELECT position,
-> COUNT(*) AS Total_Employees
-> FROM physician
-> GROUP BY position
-> ORDER BY Total_Employees DESC;
```

position	Total_Employees
Surgical Attending Physician	3
Staff Internist	1
Attending Physician	1
Senior Attending Physician	1
Head Chief of Medicine	1
MD Resident	1
Attending Psychiatrist	1

7 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM physician
-> WHERE position LIKE '%Attending%';
```

employeeid	name	position	ssn
2	Elliot Reid	Attending Physician	222222222
3	Christopher Turk	Surgical Attending Physician	333333333
4	Percival Cox	Senior Attending Physician	444444444
6	Todd Quinlan	Surgical Attending Physician	666666666
7	John Wen	Surgical Attending Physician	777777777
9	Molly Clock	Attending Psychiatrist	999999999

6 rows in set (0.00 sec)

```
mysql> SELECT p.name
-> FROM physician p
-> JOIN department d
-> ON p.employeeid=d.head
-> WHERE d.name='Surgery';
+-----+
| name |
+-----+
| John Wen |
+-----+
1 row in set (0.00 sec)

mysql> SELECT d.name,
-> p.name
-> FROM department d
-> JOIN physician p
-> ON d.head=p.employeeid
-> ORDER BY d.name;
+-----+-----+
| name | name |
+-----+-----+
| General Medicine | Percival Cox |
| Psychiatry | Molly Clock |
| Surgery | John Wen |
+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT *
-> FROM physician
-> WHERE position NOT LIKE '%Surgical%';
+-----+-----+-----+-----+
| employeeid | name | position | ssn |
+-----+-----+-----+-----+
| 1 | John Dorian | Staff Internist | 111111111 |
| 2 | Elliot Reid | Attending Physician | 222222222 |
| 4 | Percival Cox | Senior Attending Physician | 444444444 |
| 5 | Bob Kelso | Head Chief of Medicine | 555555555 |
| 8 | Keith Dudemeister | MD Resident | 888888888 |
| 9 | Molly Clock | Attending Psychiatrist | 999999999 |
+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> SELECT position,
-> COUNT(*) AS Total
-> FROM physician
-> GROUP BY position
-> HAVING COUNT(*) >= 2;
+-----+-----+
| position | Total |
+-----+-----+
| Surgical Attending Physician | 3 |
+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT *
-> FROM department d
-> WHERE EXISTS
-> (
-> SELECT 1
-> FROM physician p
-> WHERE p.employeeid=d.head
-> );
+-----+-----+-----+
| departmentid | name | head |
+-----+-----+-----+
| 1 | General Medicine | 4 |
| 2 | Surgery | 7 |
| 3 | Psychiatry | 9 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT MAX(employeeid)
-> FROM physician;
```

```
+-----+
| MAX(employeeid) |
+-----+
|                9 |
+-----+
1 row in set (0.00 sec)
```

```
mysql>
mysql> SELECT MIN(employeeid)
-> FROM physician;
```

```
+-----+
| MIN(employeeid) |
+-----+
|                1 |
+-----+
1 row in set (0.01 sec)
```

```
mysql> Show tables;
```

```
+-----+
| Tables_in_sec_13 |
+-----+
| class             |
| class_info        |
| department        |
| first_table       |
| physician         |
| second_table      |
| student           |
| student_marks1    |
+-----+
8 rows in set (0.00 sec)
```